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For a normally distributed data $x_1, x_2 \dots x_n$, the formula is,

$$x_i = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i - M)^2}{2\sigma^2}}$$

where, M = mean of data

σ^2 = variance

σ = STD

for likelihood, we'll try to fit a distribution from given set of data, unlike probability, where we try to predict probability from a given distribution.

So, for a given distribution $x_1, x_2 \dots x_n$, the likelihood for the data is from a distribution with mean M and variance σ^2

single measurement → $L(M, \sigma^2 | x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-M)^2}{2\sigma^2}}$

for all the data points →

$$L(M, \sigma^2 | x_1, x_2 \dots x_n) = L(M, \sigma | x_1) \times L(M, \sigma | x_2) \times \dots \dots L(M, \sigma | x_n)$$

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To find the maximum likelihood, we need to differentiate this function to find its peak. But calculating peak is easier if we take logarithm of this large group of products. It is to be mentioned that the peak of the function and its log occurs for the same value. So,

$$\ln L = \sum_{i=1}^{n} \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-M)^2}{2\sigma^2}}\right)$$

$$= \sum \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) + \ln e^{-\frac{(x_i-M)^2}{2\sigma^2}}$$

$$= \sum -\frac{1}{2}\ln(2\pi\sigma^2) + 2 \times -\frac{1}{2} - \frac{(x_i-M)^2}{2\sigma^2}$$

$$= \sum -\frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln\sigma^2 - \frac{(x_i-M)^2}{2\sigma^2}$$

$$= \sum_{i=1}^{n} -\frac{1}{2}\ln(2\pi) - \ln(\sigma) - \frac{(x_i-M)^2}{2\sigma^2}$$

$$\ln(L) = -\frac{n}{2}\ln(2\pi) - n\ln(\sigma) - \frac{(x_1-M)^2}{2\sigma^2} - \frac{(x_2-M)^2}{2\sigma^2} - \cdots - \frac{(x_n-M)^2}{2\sigma^2}$$

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If we take derivative to maximize this fn, we'll be able to get the values of M and σ that corresponds to maximum likelihood,

For M (mean)

$$\ln L = -n/2 \ln(2\pi) - n \ln(\sigma) - \frac{(x_1-M)^2}{2\sigma^2} + \frac{(x_2-M)^2}{2\sigma^2} \dots \frac{(x_n-M)^2}{2\sigma^2}$$

$$\frac{\partial \ln L}{\partial M} = 0 + 0 + \sum_{i=1}^{n} -1 \times 2 \times \frac{x_i - M}{2\sigma^2} \times -1$$

$$= \sum_{i=1}^{n} (x_i - M) \times \frac{1}{\sigma^2}$$

$$= \{(x_1 + x_2 \dots x_n) - nM\} \times \frac{1}{\sigma^2}$$

for this to be max

$$\{(x_1 + x_2 \dots x_n) - nM\} \times \frac{1}{\sigma^2} = 0$$

or, $x_1 + x_2 \dots x_n - nM = 0$

or, $nM = x_1 + x_2 \dots x_n$

or, $M = \frac{x_1 + x_2 \dots}{n}$ = mean of data

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for σ (standard devn)
y we can just square it for σ^2

$$\ln L = -n/2 \ln(2\pi) - n \ln(\sigma) - \sum_{i=1}^{n} \frac{(x_i - M)^2}{2\sigma^2}$$

$$\frac{\partial \ln L}{\partial \sigma} = 0 - \frac{n}{\sigma} + \frac{1}{\sigma 3} \sum_{i=1}^{n} (x_i - M)^2$$

for $\frac{\partial \ln L}{\partial \sigma} = 0$ [extreme value]

$$\frac{n}{\sigma} = \frac{1}{\sigma 3} \sum_{i=1}^{n} (x_i - M)^2$$

$$\text{or, } n = \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - M)^2$$

$$\text{or, } \sigma^2 = \frac{\sum (x_i - M)^2}{n}$$

$$\text{or, } \sigma = \sqrt{\frac{\sum (x_i - M)^2}{n}}$$

This is nothing but the standard deviation of the data points

Comparison with Gradient Descent

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Giradient descent does almost the same thing. The only difference is, the previous one was a maximizing algorithm whereas gradient descent is a minimizing algorithm. In gradient descent, we basically move opposite to the direction of the ascent to reduce certain loss function.

Here, the part upto which we calculated our derivative is the same.

$$\text{So, } \frac{\partial \ln L}{\partial M} = \sum_{i=1}^{n} (x_i - M)^2 \times \frac{1}{\sigma^2}$$

$$\frac{\partial \ln L}{\partial \sigma} = -\frac{n}{\sigma} + \sum_{i=1}^{n} (x_i - M)^2$$

So, from iteration t to next iteration $t+1$, the new estimate for M,

$$M_{t+1} = M_t - \eta \frac{\partial \ln L}{\partial M}$$, here, η = learning rate

~~for getting~~ As we approach better estimate of M, the value of the derivative term approaches zero.

At 0 value,

$$\frac{\partial \ln L}{\partial M} = 0$$

or, M = mean [same process we showed previously]

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Same logic applies of σ. As we approach better estimate of θ, the negative gradient term vanishes.

So, $\frac{\partial \ln L}{\partial \sigma} = 0$

σ = standard deviation

[same process as we showed previously]